

Rajashekar Reddy Vannavada

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with 3 years of experience building scalable, cloud-native data pipelines, and real-time analytics platforms across telecommunications, pharmaceuticals, and education domains. Skilled in AWS, Databricks, Apache Spark and modern ETL frameworks, with expertise in data governance, migrations, and analytics optimization. Proven ability to deliver business impact by reducing costs, improving data quality, and enabling domain-specific insights. Adept at collaborating with cross-functional teams to translate business requirements into scalable, secure, and cost-effective data architectures.

WORK EXPERIENCE

Data Engineer, Hillsoft Business Solutions, Remote Dec 2024 – Present

- Migrated over 500 GB of customer and billing data from legacy systems to AWS using Terraform-based IaC, ensuring GDPR/CCPA compliance and uninterrupted telecom services, reducing manual effort by 70%.
- Designed real-time streaming data pipelines using AWS Glue, Apache Spark, and kafka with Change Data Capture (CDC) to synchronize subscriber usage data, reducing migration downtime by 30% (from 8 hours to 5.6 hours).
- Created ER diagrams and hierarchical JSON models in Hackolade for data modeling in MongoDB/DocumentDB for customer device provisioning APIs, cutting integration errors by 25 percent.
- Containerized ETL jobs with Docker, improving portability dev/test/prod environments and cutting config issues by 80%.
- Optimized Redshift clusters for customer churn analytics using partitioning strategies, sort keys, and performance optimization techniques, accelerating query response time by 40%, enabling faster marketing campaign targeting.
- Deployed telecom data monitoring CloudWatch and Datadog to monitor ETL jobs and infrastructure, with Slack alerting that improved pipeline uptime to 99.9 percent and cut mean time to resolution (MTTR) from 2 hours to 45 minutes.
- Collaborated with business stakeholders to design subscriber usage insights dashboard, aligning data strategy with telecom KPIs.

Data Engineer, Takeda Pharmaceuticals, Lexington, MA Jul 2023 – Oct 2024

- Boosted Databricks ETL efficiency by 30% through Delta Lake Z-ordering and Spark performance optimization, processing 500GB/day of HR datasets across employee health and compliance systems.
- Automated Workday/ADP HR feeds into Databricks, ensuring real-time data pipelines for employee onboarding and offboarding insights for HR operations.
- Implemented Collibra-based data governance to enforce HIPAA and GDPR/CCPA, reducing regulatory risks by 25%.
- Built Delta Lake time-travel models for HR data auditing, ensuring 99.9% SLA compliance.
- Reduced storage costs by 20% using Databricks Unity Catalog and optimized Delta Lake VACUUM policies.
- Delivered Power BI dashboards for HR attrition and workforce analytics, providing 200ms latency insights.

Data Engineer, Arkansas State University, Jonesboro, AR Aug 2022 – May 2023

- Developed AWS Glue ETL jobs with PySpark to migrate over 1TB of student recruitment and campaign data into Redshift, automating data pipelines and reducing manual integration by 35%.
- Processed 500GB/day of Adobe Analytics data in near real-time student engagement insights using streaming pipelines with a latency of under 5 minutes to improve recruitment campaign targeting.
- Built CDC pipelines with DynamoDB Streams, Lambda, and S3, for student enrollment systems, achieving less than 30s data freshness for transactional updates.
- Optimized Redshift storage using Parquet formatting and partitioning strategies, reducing reporting costs by 20 percent for institutional analytics.
- Collaborated with academic departments to design self-service dashboards for student retention and enrollment forecasting.

TECHNICAL SKILLS

Programming Languages	Python (pandas, numpy, PySpark, boto3), SQL, Java, C++
Databases & Data Modeling	MySQL, PostgreSQL, DocumentDB, MongoDB, Teradata, Hackolade (ER diagrams, NoSQL schema design, data modeling)
ETL & Data Pipelines	AWS Glue, Apache Spark, Delta Lake, Kafka, Hive, CDC, Streaming
Cloud Platforms	AWS (S3, Redshift, Glue, Lambda, CloudWatch), Databricks, Snowflake
Data Governance & Security	Collibra (Data Lineage, Policy Management), GDPR, HIPAA Compliance, Unity Catalog
Data Visualization & Reporting	Tableau, Power BI, Excel
Devops & CI/CD	GitHub, Docker, Terraform, Agile, Jira
Performance Optimization	Query tuning, partitioning, vacuum automation, real-time streaming pipelines

EDUCATION

Master of Science in Computer Science, Arkansas State University, Jonesboro, AR May 2023